llmXive Automated Review of TIDE: Proactive Multi-Problem Discovery via Template-Guided Iteration

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper generally maintains a high degree of accuracy in linking its claims to the provided evidence, particularly in the qualitative case studies (Tables 1 and 2) where specific failures of baselines and successes of TIDE are meticulously detailed against the gold standard. The causal links between the proposed mechanisms (iteration, templates) and the observed metrics (coverage, precision) are well-supported by the ablation studies and scaling plots.

However, there are minor discrepancies in how specific experimental conditions are attributed to general claims in the Results section (Section 6). The text occasionally generalizes findings from a single backbone (GPT) or a specific setting (Workspace) to the broader methodology without sufficient qualification. For instance, the claim regarding the budget scaling comparison ($k = 10$ vs $k = 2$) is explicitly supported by Figure 2, which is captioned as "Workspace setting with GPT," yet the text presents the conclusion as a fundamental property of the method without reiterating the specific experimental constraints.

Similarly, the description of template counts in Section 5 is slightly ambiguous regarding whether the reported numbers (40 and 108) represent the total pool size or the per-model output, which could mislead a reader attempting to replicate the exact setup. Additionally, the description of Figure 1 in the text does not explicitly restate that these specific per-iteration curves are for the GPT backbone, potentially leading a reader to assume they represent an aggregate across all four backbones tested. While these are primarily issues of precision in reporting rather than factual errors, they require clarification to ensure the claims are strictly bounded by the evidence presented. The core scientific claims regarding the efficacy of TIDE over baselines are robustly supported by the data in Table 1 and the ablation studies.

Action items.

- [**writing**] Clarify in Section 6 whether the per-iteration coverage/precision curves in Figure 1 (fig_iteration_recall_precision) apply only to the GPT backbone or are averaged across all backbones, as the caption restricts the scope to GPT.
- [**science**] In Section 5, clarify if the template counts (40 for workspace, 108 for repo) represent the total pool size or the output per LLM, as the phrasing 'constructed by each LLM' creates ambiguity about the final pool size used in experiments.
- [**writing**] In Section 6, qualify the claim that 'Multi-Agent at k=10 falls below TIDE at k=2' by explicitly noting this specific budget comparison is limited to the GPT backbone and Workspace setting shown in Figure 2, rather than a general property.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively communicates the relationship between LLM-call budget and F1 scores across three tasks. The axes are clearly labeled with units, the legend distinguishes the methods, and the trends are easy to interpret.

Figure 2

Figure 2 is a clear and well-structured conceptual diagram that effectively illustrates the TIDE framework. The visual flow from reactive to proactive discovery and the iterative template application aligns perfectly with the provided caption, with no missing labels or confusing elements.

Figure 3

Figure 3 effectively displays per-iteration retrieval coverage and precision with clear axes, units, and a distinct legend. The data trends are easily readable, and the caption accurately describes the content shown in both subplots.

Figure 4

Figure 4 is incomplete, displaying only a legend without any corresponding data visualization or descriptive caption.

Figure 5

Figure 5 is critically flawed due to the complete absence of a caption and an unlabeled x-axis, rendering the data unintelligible. Additionally, the legend markers (dots) do not match the bar chart visualization.

Figure 6

Figure 6 effectively demonstrates the performance scaling of the 'Templates' method across three metrics, but the truncated y-axes obscure the 'Single-Agent' baseline values defined in the legend, hindering direct visual comparison.

Action items.

- **[fatal]** Figure 4: The figure contains only a legend with no actual data visualization (axes, plots, or charts) to display.
- **[fatal]** Figure 4: The caption is explicitly '(no caption)', providing no context for the legend's categories or the figure's purpose.
- **[fatal]** Figure 5: The figure has no caption, making it impossible to understand what the data represents, the meaning of the axes, or the context of the comparison.

- [**science**] Figure 5: The x-axis is completely unlabeled and lacks tick marks or values, preventing any interpretation of the distribution or the specific categories being compared.
- [**science**] Figure 5: The legend uses colored dots to represent 'GPT' and 'Gemini', but the plot consists of bar charts; the legend markers do not match the data visualization style.
- [**science**] Figure 6: The legend defines a 'Single-Agent' baseline (red line), but the y-axis scales (16-22, 12-17, 14-18) are truncated and do not show the baseline's actual value, making it impossible to visually verify the claimed performance gap.
- [**writing**] Figure 6: The y-axis labels ('Retrieval F1', 'Identification F1', 'Resolution F1') are rotated 90 degrees, which is unconventional and reduces readability compared to horizontal labels.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript exhibits a moderate overuse of domain-specific jargon and undefined acronyms that may hinder accessibility for non-specialist readers. While the core concepts are sound, the terminology often assumes familiarity with specific ML engineering conventions.

First, the term **"bottleneck"** is used extensively throughout the Method and Results sections (e.g., Section 4, Paragraph 2; Section 5, Paragraph 1; Figure 4) to describe "hidden problems" or "bugs." Although the prompt figures define a bottleneck as a "concerning issue," the main text frequently uses the term without this context, risking confusion with the standard performance engineering definition (a resource constraint). The authors should consistently use "hidden problem" or "issue" in the prose, reserving "bottleneck" only for the specific prompt definitions if necessary.

Second, the acronym **"LLM"** appears in the Abstract and Introduction without an explicit expansion at the very first instance in the Abstract. While "Large language model (LLM)" appears in the first sentence of the Introduction, the Abstract uses "LLM-based agents" (implied) or "agents" without the full expansion. The term should be explicitly defined as "Large language model (LLM)" before its first use in the Abstract.

Third, the term **"qualnames"** appears in the prompt figure captions (Figure 4, line 34) and the case study table (Table 2, line 12) without definition. This is Python-specific jargon for "qualified names." For a general audience, this should be replaced with "fully qualified function names" or "function identifiers."

Fourth, the phrase **"World Model"** is capitalized and treated as a specific input field in the Workspace setting (Figure 5, Section 5) but is not defined as a standard term in the text. This risks confusion with the broader AI concept of a "world model." The text should clarify that this refers to the "user context profile" or "state description."

Finally, the term **“gold”** is used repeatedly as an adjective (e.g., “gold problems,” “gold resolution,” “gold count”) to mean “ground truth” or “annotated reference.” While standard in machine learning, it is opaque to general readers. The authors should replace “gold” with “reference” or “annotated” in the main text (e.g., “reference problems,” “reference resolution”) to improve clarity.

Action items.

- [**writing**] The term ‘bottleneck’ is used extensively (e.g., Sections 4, 5, 6, and Figures 4-7) to describe ‘hidden problems’ or ‘bugs.’ While defined in the prompt figures, the prose often uses ‘bottleneck’ without clarifying it means ‘unarticulated problem.’ Replace with ‘hidden problem’ or ‘issue’ in the main text to avoid confusion with performance bottlenecks.
- [**writing**] The acronym ‘LLM’ is used frequently but is not explicitly defined at its first occurrence in the Abstract or Introduction. It appears in the first sentence of the Introduction as ‘Large language model (LLM) agents,’ which is acceptable, but the Abstract uses ‘LLM-based agents’ (implied) or just ‘agents’ without the expansion. Ensure ‘Large language model’ is explicitly defined before the first use of ‘LLM’ in the Abstract or Introduction.
- [**writing**] The term ‘qualnames’ appears in the prompt figure captions (e.g., Figure 4, line 34) and the case study table (Table 2, line 12) without definition. While common in Python, it is jargon for non-specialists. Replace with ‘fully qualified function names’ or ‘function identifiers’ in the text.
- [**writing**] The phrase ‘World Model’ is capitalized and used as a specific input field in the Workspace setting (Figure 5, Section 5) but is not defined as a standard term in the text. It risks confusion with the broader AI concept of a ‘world model.’ Clarify that this refers to the ‘user context profile’ or ‘state description’ in the main text.
- [**writing**] The term ‘gold’ is used repeatedly as an adjective (e.g., ‘gold problems,’ ‘gold resolution,’ ‘gold count’) to mean ‘ground truth’ or ‘annotated reference.’ This is standard in ML but opaque to general readers. Replace with ‘reference’ or ‘annotated’ in the main text (e.g., ‘reference problems,’ ‘reference resolution’).

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The logical consistency of the paper is generally strong, with a clear causal chain proposed between the problem (salient problems overshadowing subtler ones) and the solution (iterative discovery conditioned on cumulative state). The distinction between the roles of iteration (coverage) and templates (fidelity) is well-maintained throughout the results section.

However, there are two specific logical gaps regarding the internal consistency of the method’s claims versus its implementation details:

1. **The "Template-Guided" Scope Contradiction:** The Abstract and Section 4.1 assert that the framework anchors *each* prediction in a recognizable problem class via templates. However, the inference prompt in Figure 5 (`fig_prompt_inference_code`) explicitly allows the model to report issues that "do not fit any template" (setting `used_template_id` to empty string). The paper does not logically reconcile how the "template-guided" mechanism drives fidelity for these template-novel predictions. If a significant portion of high-fidelity predictions are template-novel, the claim that templates are the primary driver of fidelity (Section 6) is logically weakened. The authors should clarify if the "template-guided" label applies to the *process* of searching (using templates as priors) even when the final output is novel, or if the claim needs to be qualified.

2. **Baseline Failure Mode Assumption:** In Section 6, the paper attributes the failure of the `Multi-Agent` baseline to agents "re-anchoring on the same most-salient signal" because they lack access to each other's state. While this is a plausible hypothesis, the logical derivation assumes that independent parallel agents *will* inevitably converge on the same salient signals without any explicit control over their initialization (e.g., random seeds, temperature settings) or prompt variations. The paper treats this convergence as a necessary consequence of parallelism rather than an empirical observation of the specific baseline configuration. To strengthen the logical link, the authors should explicitly state that the parallel agents were configured to be as identical as possible (same seed, same prompt) to isolate the variable of "state sharing," or provide evidence that this convergence is a robust property of the model regardless of minor stochastic variations.

Action items.

- [**writing**] Section 4 claims templates anchor 'each prediction' in a problem class, yet Figure 5 allows 'template-novel' issues (empty ID). Clarify how the fidelity claim holds for non-template predictions or if the 'template-guided' label is misleading for those cases.
- [**science**] Section 6 attributes Multi-Agent failure to 're-anchoring' on salient signals. The paper assumes this convergence is inevitable for parallel agents without stating controls (e.g., identical seeds/temps) to isolate state-sharing as the sole variable. Explicitly confirm baseline configuration.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the generalizability and magnitude of its results that slightly exceed the immediate evidence provided in the text.

First, the Abstract and Introduction characterize the performance gains as “substantial” across all metrics and settings. While the Workspace setting shows clear and large improvements (e.g., GPT Retrieval F1: 54.32 vs 70.46), the Repository setting presents a more nuanced picture. In Table 1, the absolute F1 scores for the Repository setting are quite low (e.g., TIDE GPT Resolution F1 is 17.39, while Single-Agent is 13.27). For the Qwen backbone, the Resolution F1 improvement is from 5.76 to 9.70. While this is a relative improvement, describing it as a “substantial” gain in the context of such low absolute performance is an overstatement. The language should be calibrated to reflect that while the *relative* trend holds, the *absolute* capability in the Repository setting remains limited, particularly for the resolution component.

Second, the conclusion that “scaling parallel agents is no substitute for iterative conditioning” (Section 6) is a strong theoretical claim derived from a specific experimental comparison. The “Multi-Agent” baseline is defined as running multiple *independent* agents. This is a specific, and arguably weak, form of parallelism. A more robust baseline might involve parallel agents sharing a common scratchpad or a centralized aggregator. By framing the result as a fundamental limitation of parallel scaling rather than a limitation of *independent* parallel agents, the paper overreaches. The claim should be qualified to specify that independent parallel agents fail to match iterative conditioning, leaving open the possibility that coordinated parallelism could succeed.

Finally, the claim that templates “transfer across backbones” is supported only by data comparing GPT and Gemini (Table 2). The text implies a broader transferability that includes Claude and Qwen, for which no cross-template data is presented. To avoid overgeneralization, the authors should restrict the claim to the specific backbones tested or acknowledge the limitation.

Action items.

- [**writing**] Temper the claim of ‘substantial gains’ in the Repository setting (Abstract, Sec 2). Table 1 shows low absolute Resolution F1 scores (e.g., Qwen: 5.76 vs 9.70). Clarify that gains are relative, not necessarily substantial in absolute terms.
- [**writing**] Qualify the claim that ‘scaling parallel agents is no substitute’ (Sec 6). The baseline uses independent agents. Specify that independent parallelism fails, rather than implying all parallel scaling is inferior to iteration.
- [**writing**] Restrict the ‘templates transfer across backbones’ claim (Abstract, Sec 6). Table 2 only shows GPT-Gemini transfer. Do not imply transfer to Claude/Qwen without data.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript addresses safety and ethics primarily through a brief Ethics Statement (Section 10) and a Limitations section (Section 9). While the authors acknowledge the risks of sensitive content and bias, the current discussion is insufficient for a system designed to operate proactively without explicit user requests.

First, the **proactive intervention risk** is under-addressed. TIDE is designed to "surface hidden problems" (Abstract, Section 1) and "recommend a resolution" (Section 4). In the Personal Workspace setting, this involves identifying "bottlenecks" in a user's life based on private documents. The paper does not discuss the consequences of **false positives**—instances where the agent incorrectly identifies a non-issue as a critical bottleneck. Such errors could lead to unnecessary alarm, privacy violations (by surfacing sensitive data the user intended to keep private), or workflow disruption. The recommendation for "human-in-the-loop review" is a standard best practice but lacks specific implementation details on how the system handles the latency and friction of human verification before acting on a "hidden" problem.

Second, **data privacy and consent** regarding the evaluation datasets are unclear. Section 5.1 describes the "Personal Workspace" setting, which includes "personal pain points, relationships, and organizational roles." It is not explicitly stated whether these 30 workspaces are derived from real user data (requiring IRB approval and informed consent) or are synthetic. If real data was used, the paper must confirm that appropriate ethical review was obtained and that personally identifiable information (PII) was rigorously anonymized. If synthetic, the methodology for generating realistic yet safe synthetic data should be briefly described to ensure it does not inadvertently encode harmful stereotypes or sensitive scenarios.

Third, the **dual-use potential** in the Software Repository setting warrants more specific mitigation. The system generates "unified diff patches" (Figure 4) to fix "hidden bugs." While the intent is beneficial, an agent capable of autonomously identifying and patching code could be misused to introduce vulnerabilities or to scan proprietary codebases for exploitable flaws if the "proactive" logic is repurposed. The paper should explicitly state that the generated patches are intended for review and not for autonomous deployment, and describe any safety checks (e.g., static analysis, sandboxing) applied to the generated code before it is presented to the user.

Finally, the **bias in template construction** (Section 4) is a concern. Templates are distilled from "previously solved cases." If the training data contains historical biases (e.g., prioritizing certain types of bugs or workplace issues over others), the "thought templates" may systematically overlook or misclassify problems in underrepresented contexts. The current Ethics Statement mentions "bias detection" generally but does not specify how the authors evaluated or mitigated bias in their template library.

To improve the manuscript, the authors should expand the Ethics Statement to include specific protocols for handling false positives, clarify the provenance and consent status of the workspace datasets, and detail the safety guardrails preventing autonomous code modification.

Action items.

- [**writing**] The Ethics Statement (Section 10) and Limitations (Section 9) recommend generic safeguards (content filtering, human-in-the-loop) but lack specific protocols for the 'proactive' nature of TIDE. Explicitly address how the system prevents 'false positive' interventions where the agent might flag non-issues as critical bottlenecks, potentially causing user alarm or unnecessary workflow disruption.

- **[writing]** The Personal Workspace setting (Section 5.1) involves analyzing 'personal pain points,' 'relationships,' and 'organizational roles.' The paper does not detail the IRB approval process, consent mechanisms, or data anonymization strategies used to construct the 30 multi-problem workspaces. Clarify if these are synthetic or real user data and how privacy was preserved.
- **[writing]** The Software Repository setting (Section 5.1) uses real GitHub issues and code. While open-source, the 'proactive' generation of patches (Figure 4) carries a risk of introducing subtle bugs or security vulnerabilities if deployed without rigorous verification. The paper should explicitly state the safety guardrails preventing the agent from executing or suggesting patches that could compromise repository integrity.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The scientific evidence presented for the TIDE framework is generally robust, with a clear experimental design comparing iterative discovery against single-shot and parallel baselines across two distinct settings (Workspace and Repository). The use of four different LLM backbones strengthens the claim that the method is model-agnostic. However, several aspects of the evidence require clarification to fully support the central claims regarding fidelity and resolution quality.

First, the evaluation of "identification" and "resolution" relies entirely on an LLM judge (GPT-5 mini) using a Likert-style rubric (Section 5.3). While the authors cite G-Eval, there is no reported human evaluation or inter-annotator agreement study to validate that the LLM judge's scores correlate with human judgment. Given that the core contribution is "proactive discovery" of subtle, hidden problems, the risk of the judge hallucinating or being biased toward the method's own output (since TIDE's outputs are structured and detailed) is non-trivial. Without human validation, the claims about "fidelity" and "resolution" quality remain somewhat circular.

Second, the construction of the "Software Repository" dataset involves grouping issues from SWE-bench and TestExplora at a common anchor commit (Section 5.1). The paper does not provide a breakdown of the complexity of these grouped issues or analyze whether the number of coexisting bugs correlates with the inherent difficulty of the individual bugs. If the multi-bug instances are systematically easier or harder than single-bug instances, the observed gains in "coverage" might be confounded by dataset bias rather than the iterative mechanism itself.

Finally, the comparison with the "Multi-Agent" baseline (Section 5.2) requires more precise definition of the experimental controls. The paper states the budget k is matched, but it is unclear if the parallel agents share the same random seeds or temperature settings as the iterative rounds. Furthermore, the "scaling" analysis in Figure 5 (budget scaling) assumes a linear relationship between the number of parallel agents and the total call budget, but does not

explicitly rule out the possibility that the iterative method benefits from the cumulative context state in a way that is not directly comparable to independent parallel calls. Clarifying these experimental controls is necessary to rule out alternative explanations for the performance gap.
Action items.

- [**science**] The evaluation of 'identification' and 'resolution' relies on an LLM judge (GPT-5 mini) with a Likert-style rubric. The manuscript lacks a human-in-the-loop validation study (e.g., inter-annotator agreement or correlation with human ratings) to verify that the LLM judge's scores align with human judgment, which is critical for claims about 'fidelity' and 'resolution' quality.
- [**science**] The 'Software Repository' dataset construction groups issues from SWE-bench and TestExplora at a common anchor commit. The paper does not report the distribution of problem complexity or the correlation between the number of coexisting bugs and the difficulty of discovery, raising concerns about potential confounding variables in the multi-problem scaling analysis.
- [**science**] The 'Multi-Agent' baseline is described as running independent agents in parallel. The paper does not explicitly state whether these agents share the same random seed or temperature settings as the iterative TIDE rounds, nor does it clarify if the 'budget' k implies k total calls or k calls per agent, which is essential for a fair comparison of the 'scaling' claims in Figure 5.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in the paper is generally sound in its design of comparing TIDE against baselines, but it lacks necessary rigor in reporting uncertainty and validating the evaluation metrics.

First, the sample sizes for the evaluation splits are relatively small, particularly for the Software Repository setting (20 instances, Section 5.1). While the authors report mean Coverage and F1 scores in Table 1, they do not provide confidence intervals, standard deviations, or results of statistical significance tests (e.g., paired t-tests or Wilcoxon signed-rank tests) to confirm that the observed improvements over baselines are not due to random variance. Given the high variance often seen in LLM-based evaluations, reporting these statistics is essential to support the claim that TIDE "consistently outperforms" baselines.

Second, the reliance on an LLM judge (GPT-5 mini) for scoring Identification and Resolution (Section 5.3) introduces a potential source of bias and inconsistency that is not quantified. The paper does not report inter-rater reliability metrics (such as Cohen's Kappa) between the LLM judge and human annotators, nor does it provide a validation study demonstrating the judge's consistency across different prompts or instances. Without this validation, the absolute scores and the magnitude of the gains reported in Table 1 are difficult to interpret with confidence.

Finally, the scaling analyses presented in Figure 3 (budget scaling) and Figure 5 (template pool size) lack error bars or confidence intervals. These figures are used to argue for the effectiveness of iterative discovery and template scaling, but without visual or statistical representation of variance, it is unclear if the observed trends are robust. The authors should re-run these experiments with multiple seeds or report the variance across the existing runs to substantiate these claims.

Action items.

- [**science**] Section 5.3 states metrics are 'macro-averaged across instances' but does not report confidence intervals or standard errors for the main results in Table 1. Given the small sample size in the Repository setting (20 instances), statistical significance testing (e.g., paired t-tests or Wilcoxon signed-rank) is required to validate the claimed gains over baselines.
- [**science**] The LLM judge (GPT-5 mini) is used for Identification and Resolution scoring (Section 5.3). The paper lacks an inter-rater reliability analysis (e.g., Cohen's Kappa) or a human-in-the-loop validation study to establish the validity and consistency of these automated metrics, which are central to the paper's conclusions.
- [**science**] Figure 3 (budget_scaling) and Figure 5 (template_count_scaling) show performance trends but lack error bars or confidence intervals. Without these, it is impossible to determine if the observed scaling effects are statistically significant or within the noise of the evaluation process.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high standard of academic writing, with a clear logical flow and precise vocabulary that effectively communicates the novelty of the TIDE framework. The abstract and introduction successfully frame the problem of "discovering multiple hidden problems from context," and the transition from the limitations of reactive agents to the proposed iterative solution is smooth and compelling. The prose is generally concise, avoiding unnecessary jargon while maintaining technical rigor.

However, there are a few minor grammatical and stylistic issues that, while not obscuring the meaning, detract slightly from the overall polish. In Section 2, the sentence "These otherwise different issues share a common structure that extends beyond the workspace setting above" creates a momentary ambiguity regarding what "structure" refers to; a slight rephrasing to explicitly link the structure to the "multi-problem coexistence" pattern would improve clarity. Additionally, in Section 4, the phrase "where none of which is articulated" contains a grammatical error regarding the relative pronoun and verb agreement, which should be corrected to "none of which are articulated" or restructured for better flow. Finally, in Section 6, the use of inline mathematical notation like "k=10" within a narrative sentence breaks the reading rhythm; converting these to prose (e.g., "a budget of 10") would enhance readability without losing precision. Addressing these small points will further elevate the quality of the manuscript.

Action items.

- [**writing**] In Section 2 (Introduction), the sentence 'These otherwise different issues share a common structure that extends beyond the workspace setting above' is slightly ambiguous. Clarify that the 'structure' refers to the multi-problem coexistence pattern described in the preceding paragraph, not the issues themselves.
- [**writing**] In Section 4 (Method), the phrase 'where none of which is articulated' in the task formulation paragraph contains a grammatical error. It should be corrected to 'none of which are articulated' or 'where no problem is articulated' to ensure subject-verb agreement and clarity.
- [**writing**] In Section 6 (Results), the sentence 'Interestingly, Multi-Agent at k=10 still falls below TIDE at k=2' uses a mathematical notation style that interrupts the prose flow. Consider rephrasing to 'Interestingly, the Multi-Agent baseline with a budget of 10 still underperforms TIDE with a budget of 2' for better readability.